

Support, Readout, and the Binary-Rank Obstruction

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Abstract

Operational reconstructions in algebraic quantum field theory and quantum information reconstruct the algebraic object that is supplied: a represented von Neumann algebra, a locally covariant functor, a sector category, a correctable logical algebra, or a Type II algebra with a specified trace and entropy normalization. This note isolates a finite central obstruction to turning such a reconstruction into a derivation of a binary microscopic alphabet. If the displayed data factor through a normal quotient $\pi_q : \widehat{\mathcal{A}}_q \rightarrow \mathcal{A}$, where $\widehat{\mathcal{A}}_q = \mathcal{A} \oplus (\mathcal{N} \bar{\otimes} \ell^\infty(\{1, \dots, q\}))$, then the same data are compatible with $q = 2$ and with $q > 2$ finite central completions. They therefore do not determine full ambient support, faithfulness of the represented physical algebra, the physical/null/gauge/ inadmissible status of $\ker \pi_q$, the finite central rank q , or the predicate $q = 2$.

The theorem also records the exact positive form of the missing hypotheses. First, every non-null finite central direction must be accounted for by available probes/readouts or by a null, gauge, or inadmissibility quotient: for $V_q = \ell_q^\infty / \mathbb{C}1$, this is the rank equation $\text{rank } R_q = \dim(V_q / N_q)$. Second, even after support/readout accountability, binary rank does not follow from entropy alone; every $q \geq 2$ satisfies $S_q = \log q = \Delta A_q / (4G\hbar)$ after choosing $\Delta A_q = 4G\hbar \log q$. A binary conclusion requires a further finite-alphabet law, such as the least nontrivial closed quotient rule, or direct $q = 2$ spectroscopy whose measurement channel is itself derived. A small Lean 4 file verifies the finite same-data/no-reconstruction kernel.

Keywords. Algebraic quantum field theory; von Neumann algebras; finite centers; support projections; locally covariant quantum field theory; DHR sectors; operator-algebra quantum error correction; black-hole spectroscopy.

Status. This is a submission-freeze package extracted from existing notes. It does not modify the frozen finite-center paper. Its claim is intentionally negative and conditional: quotient-visible data do not force a binary primitive.

1 Question

The intended target is not a new reconstruction theorem for AQFT, modular theory, entanglement-wedge reconstruction, or gravitational crossed products. Those positive theorems are accepted as inputs. The question is a narrower converse. Suppose an operational experiment supplies normal state and effect values, channels, modular data, locally covariant time-slice and relative Cauchy evolution data, DHR/DR sector data, JLMS/OAQEC recovery data, continuum limits, and entropy assignments, but all of the displayed data factor through a represented quotient $\mathcal{A}$. Can those data alone prove that there is no additional finite central support in a larger admitted algebra $\widehat{\mathcal{A}}_q$? Can they decide whether such a finite central kernel is physical, null, gauge, or inadmissible? Can they select the special finite rank $q = 2$?

The answer in this note is negative under explicit assumptions. The same answer is also the useful positive guide: a proof of binary rank must stop using quotient-visible data and must exhibit the extra non-quotient input. In finite language, that input has two independent pieces. The first is support/readout accountability: all finite central directions must either be separated by physical probes/readouts or be removed by a null/gauge/ inadmissibility rule. The second is a primitive finite-alphabet rule: after a finite center is accountable, entropy and ordinary yes/no tests still do not tell us which quotient is counted as the elementary finite alphabet.

2 Novelty Boundary

The table gives the novelty check. The note should be demoted if a published theorem already states the joint criterion in the right form.

Claim in this note	Nearest standard result	Honest boundary
Operational data reconstruct only targets constant on data fibers.	This is elementary descent for functions, and in operator algebras it appears whenever a normal quotient or represented null space is fixed.	No novelty is claimed for the fiber lemma. The contribution is the finite central application to support, readout, and binary rank.
Quotient-visible locally covariant and relative-Cauchy data for a supplied functor do not decide whether an invisible finite center has been admitted.	Brunetti–Fredenhagen–Verch define locally covariant QFT and relative Cauchy evolution; Fewster–Verch define dynamical locality and SPASs [1, 2].	No new locally covariant theorem is claimed. Full-theory dynamical locality, source faithfulness, or indecomposability may be exactly the extra positive assumption; quotient-visible data alone are weaker.

DHR/DR sector data do not recreate central-character fibers omitted from the supplied category.	Doplicher–Haag–Roberts and Doplicher–Roberts reconstruct field algebras and gauge groups from a supplied sector structure [3, 4, 5].	No new DR reconstruction theorem is claimed. The statement is a warning about which category was supplied.
JLMS/OAQEC recovers a supplied logical algebra, not a larger finite central completion.	JLMS, Harlow, Dong–Harlow–Wall, and infinite-dimensional OAQEC work identify relative entropy/recovery conditions for specified algebras [6, 7, 8, 9].	The note does not dispute recovery. It says maximality or full logical algebra data is an additional input.
Crossed-product/generalized-entropy data do not select $q = 2$ without finite rank calibration.	Witten, Chandrasekaran–Longo–Penington–Witten, and Chandrasekaran–Penington–Witten construct Type II algebras and entropy notions in gravitational settings [10, 11, 12].	The note does not rederive generalized entropy. It isolates finite residuals when a central register is outside the supplied algebra or trace calibration.
Black-hole entropy does not by itself force binary rank.	Bekenstein–Mukhanov black-hole spectroscopy relates area spacing, degeneracy, and line emission [13].	A resolved $\log 2$ line would validate binary empirically. The note says entropy alone is not that measurement.

3 Data and Targets

Let $\mathcal{A}$ be the represented algebra reconstructed from the displayed operational experiment. Let $\mathcal{N}$ be a nonzero von Neumann algebra and, for each $q \geq 2$, set

$$\widehat{\mathcal{A}}_q = \mathcal{A} \oplus (\mathcal{N} \bar{\otimes} \ell^\infty(\{1, \dots, q\})).$$

Let $\pi_q : \widehat{\mathcal{A}}_q \rightarrow \mathcal{A}$ be the normal quotient onto the first summand. The finite central projection

$$p_q = (0, 1_{\mathcal{N} \bar{\otimes} \ell_q^\infty})$$

is nonzero in $\widehat{\mathcal{A}}_q$ and is killed by π_q .

The displayed data are allowed to be rich, but in the negative theorem they are assumed to factor through π_q . Examples include:

- pulled-back normal state/effect values $a \mapsto \varphi(\pi_q(a))$;
- represented modular data for a weight after quotienting to $\mathcal{A}$;
- Osterwalder–Schrader or GNS data after the represented null quotient;

- a locally covariant functor, time-slice map, or relative Cauchy evolution computed only for the quotient algebra;
- a supplied DHR/DR sector category for $\mathcal{A}$;
- JLMS/OAQEC relative-entropy or recovery data for a specified logical algebra contained in $\mathcal{A}$;
- a continuum limit and a fixed entropy value computed only in $\mathcal{A}$.

The targets are ambient. They include full support in $\widehat{\mathcal{A}}_q$, faithfulness of the represented physical algebra on $\widehat{\mathcal{A}}_q$, the interpretation of $\ker \pi_q$ as physical/null/gauge/inadmissible/absent, the finite rank q , and the predicate $q = 2$. If the target is only the represented quotient $\mathcal{A}$, the negative theorem is irrelevant: standard positive reconstruction theorems apply to the object actually supplied.

4 Main Classification

Theorem 1 (support/readout and binary-rank obstruction). *Let the comparison class contain the algebras*

$$\widehat{\mathcal{A}}_q = \mathcal{A} \oplus (\mathcal{N} \bar{\otimes} \ell^\infty(\{1, \dots, q\}))$$

for at least two values $q \geq 2$, including $q = 2$ and some $q > 2$. Suppose that every displayed datum D_q is a labelled function of the normal quotient $\pi_q : \widehat{\mathcal{A}}_q \rightarrow \mathcal{A}$, and suppose no additional datum separates the second summand or removes it from the comparison class by a quotient, gauge, inadmissibility, factor, split, or physical-exclusion rule.

Then all displayed data are identical as q varies. Nevertheless $p_q = (0, 1)$ is a nonzero central projection killed by all quotient-visible readouts. Consequently the data do not reconstruct:

1. full ambient support in $\widehat{\mathcal{A}}_q$;
2. faithfulness on $\widehat{\mathcal{A}}_q$;
3. whether $\ker \pi_q$ is physical, null, gauge, inadmissible, or absent;
4. the finite central rank q ;
5. the binary predicate $q = 2$.

Moreover, if the finite central directions that survive quotienting are encoded by $V_q = \ell_q^\infty / \mathbb{C}1$, if $N_q \subset V_q$ is the declared null/gauge/inadmissible subspace, and if $R_q : V_q \rightarrow \mathbb{R}^m$ is the available readout map, then full finite-central accountability is exactly the rank condition

$$\text{rank } R_q = \dim(V_q/N_q),$$

with R_q understood on V_q/N_q . If this accountability condition is not met, there are non-null finite central directions that are not read out. If it is met, binary rank still does not follow from the Bekenstein–Hawking coefficient alone: for every $q \geq 2$,

$$S_q = \log q, \quad \Delta A_q = 4G\hbar \log q$$

satisfy $S_q = \Delta A_q/(4G\hbar)$. A binary conclusion therefore requires an additional finite-alphabet rule selecting the least nontrivial closed quotient, or direct $q = 2$ data with its measurement channel.

Proof. For the first part, every displayed datum has the form $\tilde{D} \circ \pi_q$. Hence it is independent of the second summand and is the same for all q . The projection p_q is central and nonzero in $\hat{\mathcal{A}}_q$, but $\pi_q(p_q) = 0$, so all pulled-back normal readouts and all quotient-formed modular, sector, recovery, continuum, and entropy data vanish on the corner $p_q \hat{\mathcal{A}}_q p_q$. Replacing $q = 2$ by $q = 3$ keeps all displayed data fixed while changing the finite central rank and the truth value of $q = 2$. Any rule depending only on the displayed data would assign the same answer to both candidates, so no such rule can reconstruct the listed ambient targets.

The accountability statement is finite linear algebra. After quotienting by the declared null/gauge/inadmissible directions N_q , the remaining finite central directions are V_q/N_q . A family of real-valued central readouts separates all of them exactly when its induced map on V_q/N_q is injective. For finite-dimensional spaces, injectivity is equivalent to rank $R_q = \dim(V_q/N_q)$. If the rank is smaller, a nonzero class is both non-null and invisible to the supplied readouts.

Finally, the entropy statement is again finite. For an accountable q -state event alphabet, the Shannon/von Neumann entropy of the uniform finite cell is $\log q$. Choosing the corresponding area increment $\Delta A_q = 4G\hbar \log q$ gives the exact one-quarter coefficient for every integer $q \geq 2$. Thus the coefficient evaluates the alphabet once the alphabet has been specified; it does not choose the alphabet. Choosing the least nontrivial quotient is a separate finite-alphabet rule, and a resolved $\log 2$ spectral line is a direct datum rather than a consequence of the quotient-visible data. $\square$

5 The Countermodel Family in Coordinates

It is useful to spell out the same-data family without any gravitational language. Fix a represented algebra $\mathcal{A}$, a nonzero algebra $\mathcal{N}$, and a finite rank $q \geq 2$. The enlarged candidate is

$$\hat{\mathcal{A}}_q = \mathcal{A} \oplus (\mathcal{N} \bar{\otimes} \ell_q^\infty).$$

The quotient-visible experiment uses the map

$$\pi_q(a, n_1, \dots, n_q) = a.$$

Every normal state ω on $\mathcal{A}$ gives a pulled-back normal state $\omega \circ \pi_q$ on $\hat{\mathcal{A}}_q$. Every effect, channel matrix element, relative entropy, modular expectation value, or continuum

observable formed from the quotient is similarly pulled back. Thus the entire displayed experiment is independent of q .

The ambient algebra, however, is not independent of q . Its center contains the projection $p_q = (0, 1, \dots, 1)$, and the finite central quotient of the hidden summand has q characters. A rank-sensitive target can distinguish $\widehat{\mathcal{A}}_2$ from $\widehat{\mathcal{A}}_3$, while the displayed data cannot. This is the whole no-go in its shortest form: same data, different target. The argument is insensitive to whether $\mathcal{N}$ is type I, type II, or type III; if the finite central summand is admitted and not separated or quotiented, it survives the displayed quotient data.

There are two common ways to misread this example. The first is to say that a universal representation of the full algebra would see the finite center. That is true, but it changes the data. The universal representation of $\widehat{\mathcal{A}}_q$ contains states supported on the hidden summand, while the quotient-visible experiment does not. The second is to say that the hidden summand is obviously unphysical. That may also be true in a better theory, but then the better theory must state the physicality, quotient, gauge, or inadmissibility rule that removes it. The point of the countermodel is that the quotient-visible data themselves do not provide that rule.

5.1 Positive-support readouts versus atom separation

For a specified finite center $C_q = \ell_q^\infty$, a family of positive normal readouts can be faithful on the positive cone without separating the atoms. In C_2 , the sum readout

$$\tau(z_1, z_2) = z_1 + z_2$$

is strictly positive on every nonzero positive vector, so it detects whether a positive element is zero. But it does not distinguish $(1, 0)$ from $(0, 1)$, and its kernel is not an algebra ideal. Thus positive support faithfulness and full central tomography are different exits. The first rules out invisible positive support relative to the specified center; the second reconstructs atoms, central weights, and finite rank. A binary-rank theorem needs the second kind of information, or a physical law replacing it.

5.2 Static finite labels and response data

The same finite central extension can be made static with respect to the quotient response. If a locally covariant or Hamiltonian response is computed only on $\mathcal{A}$, take the hidden central register to have identity evolution and zero source response in the displayed experiment. Then every quotient response function is unchanged. This does not prove that response-null finite labels are physically admissible. It says only that response-null labels are not excluded by response data unless the theory also asserts response-faithful probe admissibility, source-generated physicality, or a null/inadmissibility quotient for response-null finite directions.

5.3 All binary questions do not make a binary alphabet

For $q > 2$, the finite algebra C_q has many two-outcome coarse-grained tests:

$$(\chi_A, 1 - \chi_A), \quad \emptyset \neq A \neq \{1, \dots, q\}.$$

Owning these yes/no questions is not the same as declaring one of them to be the elementary formation alphabet. A three-outcome register has three binary questions and still has three atoms. A four-outcome register may be a pair of bits, a single four-letter alphabet, or a coarse quotient of a larger alphabet, depending on which tensor/refinement structure the physical theory supplies. Thus binary propositions are cheap; binary primitive rank is not.

6 What Would Falsify the No-Go

The theorem is not meant to be immune to stronger hypotheses. It is designed to make those hypotheses explicit. Any of the following would block the countermodel family.

Full central tomography. If the experiment includes normal preparations and effects separating all minimal central projections of $\hat{\mathcal{A}}_q$, then q is measured. The same-data fiber collapses. This is not a contradiction of the theorem because the data no longer factor through π_q .

A quotient or physicality theorem. If a theorem says that every finite central direction outside the represented GNS/operational support is null, gauge, or inadmissible, then the hidden summand is removed before ambient support is demanded. Again, the theorem is not contradicted; one of its excluded positive exits has been supplied.

Maximality of the logical algebra. If OAQEC/JLMS data are not merely data of a correctable subalgebra but data of the full maximal logical algebra, then a hidden finite center in the logical commutant is no longer invisible. The missing input is the maximality statement, not recovery of the supplied subalgebra.

A source-faithfulness principle. If all admitted finite labels must have nonzero source, stress, modular, or relative-Cauchy response unless they are null, then response-null q -labels are inadmissible. This is a strong physical admissibility condition. It is plausible in some settings and false in others with transparent topological or superselection sectors; the note treats it as an assumption that must be proved for the class under discussion.

A binary finite-alphabet law. If every accountable finite event alphabet is required to be the least nontrivial closed quotient of its event algebra, then the selected size is 2. The Lean core checks this finite implication. What remains is the physical derivation of the least-quotient rule.

Direct spectroscopy. If black-hole spectroscopy or another finite-rank probe resolves $\Xi = \log q$, and the measurement channel is part of the microscopic theory, then the observed q is selected. A result $\Xi = \log 2$ validates binary; $\Xi = \log 3$ or $\Xi = \log 4$ falsifies it. This route is empirical rather than a derivation from quotient-visible data.

7 Referee Checklist

A referee can check the note without adopting any private formalism by asking five questions.

1. Is the comparison class explicitly closed under the finite central direct sums used in the countermodel? If not, where is the physical exclusion theorem?
2. Do the displayed data genuinely factor through the quotient π_q ? If some datum does not factor through π_q , does it separate the hidden central rank?
3. Is the target ambient rather than quotient-internal? If the target is only the represented algebra $\mathcal{A}$, the no-go is irrelevant.
4. Is support/readout accountability supplied by full central tomography, by a null/gauge/inadmissibility quotient, or by the rank equation $\text{rank } R_q = \dim(V_q/N_q)$?
5. After accountability, what selects $q = 2$ rather than $q > 2$: a least-event law, an observed $\log 2$ spectral datum, or some other rank-breaking microscopic principle?

If the answer to the fourth or fifth question is missing, the binary conclusion is conditional. If all five questions have sharp answers, the note provides a map for where the positive proof should attach.

8 How the Standard Reconstructions Enter

8.1 Modular and crossed-product data

Let M_0 and M_1 be von Neumann algebras with faithful normal semifinite weights ϕ_0 and ϕ_1 . For

$$M = M_0 \oplus M_1, \quad \phi = \phi_0 \oplus \phi_1,$$

the modular automorphism group decomposes componentwise:

$$\sigma_t^\phi = \sigma_t^{\phi_0} \oplus \sigma_t^{\phi_1}.$$

Accordingly, the crossed product decomposes as

$$M \rtimes_{\sigma^\phi} \mathbb{R} \cong (M_0 \rtimes_{\sigma^{\phi_0}} \mathbb{R}) \oplus (M_1 \rtimes_{\sigma^{\phi_1}} \mathbb{R}).$$

If the supplied modular or Type II data are formed after quotienting to M_0 , they are data of M_0 . They do not recover M_1 , its finite central rank, or its kernel classification. The full

direct-sum core would see the second summand, but supplying that full core is precisely an additional non-quotient datum. This is compatible with, and does not weaken, the crossed-product entropy program: the entropy is an entropy of the algebra and trace actually supplied.

8.2 Locally covariant QFT and relative Cauchy evolution

In the Brunetti–Fredenhagen–Verch framework a theory is a functor from globally hyperbolic spacetimes to algebras, and relative Cauchy evolution measures the response of that supplied functor to metric perturbations. A constant finite central extension

$$\widehat{\mathcal{A}}_q(M) = \mathcal{A}(M) \oplus (\mathcal{N}(M) \bar{\otimes} \ell_q^\infty)$$

can be arranged so that the quotient-visible functor, its time-slice maps, and its relative Cauchy evolution are exactly those of $\mathcal{A}$. The finite central summand has zero quotient response because the quotient never asks it a question. This paragraph is not a counterexample to Fewster–Verch dynamical locality for a fully specified theory. It says only that the relative-Cauchy and functorial records of the represented quotient do not decide which full theory was admitted. Local covariance and dynamical locality close the gap only when imposed on the full admitted theory, or when an independent indecomposability, source-generated closure, source faithfulness, or operational physicality rule forbids the invisible extension. They do not, as quotient data alone, select $q = 2$.

8.3 DHR/DR sector data

DHR and Doplicher–Roberts reconstruction start from a specified observable net and a specified category of localized transportable endomorphisms. If the supplied category is the category for $\mathcal{A}$, and the q central characters in an enlarged algebra are all restricted to the same object of that category, then DR reconstruction reconstructs the field algebra and compact gauge group associated to the supplied category. It does not recreate central-character fibers that were omitted from the sector datum. The repair is also exact: supply the amplified category $\mathcal{S}(\mathcal{A}) \times \{1, \dots, q\}$, or supply another datum that separates the central characters, or prove that the central-character fibers are null, gauge, or inadmissible.

8.4 JLMS and operator-algebra quantum error correction

Operator-algebra quantum error correction is intentionally algebra-relative. Exact recovery of an algebra $\mathcal{M}$ through a channel says that observables in $\mathcal{M}$ can be represented on the recovery region. It is downward closed: subalgebras of a correctable algebra are correctable. Therefore exact recovery of a diagonal or quotient algebra does not imply that the algebra is maximal among all admitted finite central extensions. JLMS relative-entropy equality has the same support: it identifies the relative entropy of the supplied

bulk and boundary algebras. To force the finite central rank, one must know that the supplied algebra is the full logical algebra, or else supply central tomography, central-sector calibration, or a quotient/inadmissibility rule.

8.5 Entropy and black-hole spectroscopy

The Bekenstein–Hawking coefficient fixes the ratio between a chosen area increment and the entropy of the chosen finite alphabet. It does not choose the alphabet. For every finite $q \geq 2$, the pair

$$(S_q, \Delta A_q) = (\log q, 4G\hbar \log q)$$

has the exact one-quarter coefficient. Bekenstein–Mukhanov spectroscopy is therefore the natural empirical boundary: a measured line spacing $\log q$, with the measurement channel and propagation model under control, would select the observed q . A $\log 2$ observation would validate the binary branch. But that would be a new q -sensitive datum, not a derivation from quotient-visible entropy/OS/continuum data.

9 Positive Exits

The theorem is useful only if it says exactly what would close the obstruction. The current exits are:

Exit	What it proves
Central support faithfulness	No nonzero positive finite central projection remains outside the represented support.
Full ambient central tomography	Minimal central projections, central weights, the finite rank q , and the predicate $q = 2$ are separated by supplied readouts.
Null/gauge/inadmissibility classifier	Every invisible finite central direction is removed before full support is demanded.
Full logical/maximal correctable algebra data	Recovery/JLMS data are known to refer to the full admitted logical algebra, not only to a downward-closed recovered subalgebra.
Full locally covariant/source-generated physicality	Response-null finite labels are operationally null or inadmissible, rather than untested physical sectors.
Support/readout accountability rank	For $V_q = \ell_q^\infty/\mathbb{C}1$, the available readout rank equals $\dim(V_q/N_q)$.
Binary finite-alphabet law	After accountability, the elementary alphabet is the least nontrivial closed quotient, or direct $q = 2$ data with a derived measurement channel is supplied.

These exits are not equivalent. Support/readout accountability tells us that all surviving finite central directions are either read out or removed; it does not by itself say that the elementary event alphabet has two outcomes. The binary finite-alphabet law tells us how to count an accountable finite event algebra; it does not by itself prove that all physical finite central directions have been accounted for. A successful positive derivation of binary rank must supply both.

10 Primitive Counting After Accountability

Assume, for this section only, that the finite support problem has been solved: the q -cell lies in the physical algebra, its central idempotents are readable or null-quotiented, fixed-area entropy counts operationally preparable states, and remnant-free recovery preserves the relevant input distinctions. The binary problem still remains.

Let $C_q = \ell^\infty(\{1, \dots, q\})$, and suppose all binary coarse-grained tests $(\chi_A, 1 - \chi_A)$ are available for nonempty proper $A \subset \{1, \dots, q\}$. This does not imply that the elementary

formation alphabet has two elements. For $q = 3$, $q = 4$, or $q = 8$, all such binary questions can exist while the complete q -register remains the elementary alphabet. Landauer cost, Blackwell/Petz sufficiency, and the area law evaluate or compare the alphabet after the experiment has been declared; they do not choose the experiment.

Thus a finite counting functional is needed. Three examples are:

$$\begin{aligned} \text{least nontrivial quotient} &\mapsto 2, \\ \text{complete register} &\mapsto q, \\ \text{selected } k\text{-block quotient} &\mapsto k. \end{aligned}$$

The first rule forces binary. The second and third do not. The question for a microscopic theory is therefore not whether yes/no propositions can be formed inside a finite event algebra; they can. The question is why the least nontrivial quotient, rather than the complete finite register or another coarse quotient, is the elementary counted alphabet. In the present note this is an assumption to be derived or measured, not a theorem.

11 Binary Blocks Versus Elementary Binary Rank

A recurring ambiguity is the difference between representing a finite alphabet with binary variables and proving that the elementary physical alphabet has two outcomes. These are not the same statement.

If $q = 2^m$, then C_q is abstractly isomorphic to the algebra of functions on m classical bits. This observation is useful only after the theory has also supplied the corresponding tensor/refinement structure as physical. An abstract isomorphism does not say that the m bit coordinates are available as independently preparable, independently readable, or independently counted subevents. Without that extra structure, the complete q -letter register can remain the elementary alphabet even though it admits a binary coding.

If q is not a power of two, binary source coding embeds the q -letter alphabet into a larger binary code space with unused words. That again does not derive binary rank. It replaces one physical question by another: why are the unused code words inadmissible, null, or gauge, and why is the larger binary code space the physical one rather than the original q -letter alphabet? The embedding is a representation of finite information, not a classification of microscopic event atoms.

This distinction matters for $q = 4$ and $q = 8$. A positive binary theory may well reclassify them as composite binary blocks, but it can do so only when it owns the binary factorization. If the physical event algebra is merely a symmetric four-letter center with no distinguished coordinate partitions, then there is no invariant choice of two bit coordinates. Choosing one is a symmetry-breaking or refinement datum. Such a datum may be physical, but it is not supplied by the quotient-visible data.

Prime $q > 2$ is sharper. A prime three-letter or five-letter accountable register cannot be decomposed into smaller nontrivial Cartesian alphabets. For those cases, a binary conclusion must either exclude the register, prove it null/gauge/inadmissible, observe a different q , or supply a rule saying that the elementary event is a two-outcome quotient

rather than the complete prime register. The least nontrivial closed quotient rule does exactly that, but it is the rule to be derived.

The same issue appears in statistical sufficiency language. Minimal sufficient statistics are minimal relative to a specified family of future experiments. If the future experiment distinguishes all q outcomes, the full q -statistic is minimal. If the future experiment asks only a fixed binary question, a binary statistic is sufficient. Sufficiency theory therefore organizes the experiment once the experiment is declared. It does not by itself declare which finite experiment is the microscopic formation event.

Landauer erasure has the same limitation. Erasing a k -letter classical alphabet at temperature T costs $k_B T \log k$ in the ideal reversible bound. The formula evaluates the alphabet. It does not say whether the alphabet was 2, 3, 4, or a binary block decomposition of 4. In the black-hole application, replacing $\log k$ by $\Delta A_k / (4G\hbar)$ preserves the same structure: the one-quarter coefficient fixes the conversion once k is known.

Thus Paper 2 should not try to prove binary by pointing to bits as a universal language for finite information. It should prove, assume, or observe one of three rank-breaking inputs:

1. a physical exclusion of every accountable nonbinary elementary register;
2. a physical factorization/refinement theorem that demotes composite 2^m -registers to binary blocks and excludes or quotients prime $q > 2$;
3. direct q -sensitive data whose result is $q = 2$.

Absent one of these, a binary encoding is notation, not a microscopic derivation.

12 Assumptions Not Proved

Assumption	Status in this note
The comparison class admits finite central completions $\widehat{\mathcal{A}}_q$ for more than one q .	This is the countermodel class. A physical theory may forbid it, but that requires an explicit exclusion, quotient, factor, split, or physicality rule.
Displayed data factor through the represented quotient.	This is the negative theorem's hypothesis. A positive theorem must show that the data do not factor through π_q .
The target is ambient support, kernel status, finite rank, or binary rank.	If the target is only the represented quotient, ordinary positive reconstruction theorems are untouched.
Support/readout accountability is physically forced.	Not proved. The note only states the finite rank equation that would express it.

Operational entropy counts exactly preparable finite support.	Not proved. If entropy counts unpreparable labels, Page/recovery arguments do not see them.
The elementary finite alphabet is the least nontrivial closed quotient.	Not proved. This is the binary-counting law.
Direct $q = 2$ spectroscopy exists and its measurement channel is derived.	Not supplied. Such data would validate binary empirically but would not be a derivation from quotient-visible data.

13 Lean Core

The accompanying Lean archive

`docs/support_readout_q2_lean_verification/`

checks only the finite logical core. It imports Lean’s `Std` library and proves:

- a same-data pair with different target values forbids reconstruction from the data;
- the same statement for a proposition-valued target;
- the blind $q = 2$ and $q = 3$ candidates have the same data but different rank and different truth values of “rank is two”;
- a missing centered readout/null rank prevents accountability;
- if the least nontrivial event-size rule is supplied for $q \geq 2$, it selects size 2.

It does not formalize von Neumann algebras, DHR theory, locally covariant QFT, JLMS, crossed products, or black-hole spectroscopy. Those are literature inputs to the manuscript. The Lean file is therefore evidence for the finite descent kernel, not evidence for the analytic or physical assumptions.

14 Conclusion

The support/readout obstruction is not a failure of AQFT or quantum information reconstruction. It is a boundary on what the displayed data are allowed to imply. If all data factor through a represented quotient, they cannot decide the status or rank of an invisible finite central kernel. If the finite center is made accountable, entropy still does not choose $q = 2$; binary rank needs a finite-alphabet law or direct binary data.

This turns the remaining binary problem into a short checklist. A positive theory must either derive support/readout accountability and then derive the least-nontrivial-alphabet law, or it must produce a q -sensitive measurement whose channel is itself part of the theory and whose outcome is $\log 2$. Without one of those routes, the honest conclusion is conditional framework, not primitive unavoidability.

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